MATH 104 FINAL
SUMMER 2016

Questions	1	2	3	4	5	6	Total
Points							
Out of	10	10	20	20	20	20	100

(Leave this table empty. For instructor's use only.)

Student Number	*Last Name*	*First Name*

Duration: 2 hours

Rules:

- Read, sign and observe the honor code below.
- **In all of the questions in this exam, *explain* your answers by showing the steps of the derivation in a *clear, organized* fashion. You might get *no* points for correct answers with no derivation and/or no explanation and/or unintelligible/unorganized solution.**
- No calculators or any other electronic equipment is allowed.
- You may use both sides of the paper.
- Respect others by being silent. Talking will not be tolerated! You risk being *thrown out of the exam* if you talk even about things not related to questions. If you have a question, silently raise your hand when one of the instructors is in the class.
- Leaving classroom in the middle of the exam is not allowed. You can submit your sheet and leave early if you want.
- If a student displays inappropriate behavior during the exam, the proctors may stop testing and report the inappropriate behavior to the Dean's Office.

Honor Code:
I pledge under my honor that I **did not give or receive any unauthorized assistance** in this work.

Signed:.....

1. [4 + 4 + 2 = 10 pts] **(Curves in space)** A wall lizard is moving along a spiral, where the position vector is given by $\vec{\mathbf{r}}(t) = \underbrace{t\cos t}_{x(t)}\,\hat{\mathbf{i}} + \underbrace{t\sin t}_{y(t)}\,\hat{\mathbf{j}} + \underbrace{(9-t^2)}_{z(t)}\,\hat{\mathbf{k}}$ for $0 \le t \le 2$, where t denotes time.

(i) Show that the lizard is moving on the paraboloid $z = 9 - x^2 - y^2$.

$$9 - [x(t)]^2 - [y(t)]^2 = 9 - [t\cos t]^2 - [t\sin t]^2$$
$$= 9 - t^2\underbrace{(\cos^2 t + \sin^2 t)}_{1} = 9 - t^2 = z(t) \checkmark$$

(ii) Find the speed $|\vec{\mathbf{v}}|$ of the wall lizard (as a function of time).

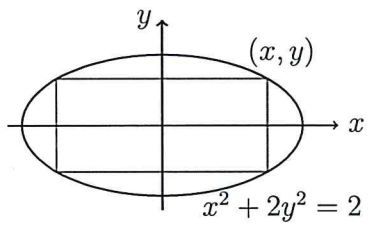
$$\vec{v}(t) = (\cos t - t\sin t)\hat{i} + (\sin t + t\cos t)\hat{j} - 2t\hat{k}$$

$$|\vec{v}| = \sqrt{\boxed{\cos^2 t} + t^2\boxed{\sin^2 t} - \cancel{2t\cos t\sin t} + \boxed{\sin^2 t} + t^2\boxed{\cos^2 t} + \cancel{2t\cos t\sin t} + 4t^2}$$
$$= \sqrt{1 + t^2 + 4t^2} = \sqrt{1+5t^2}$$

(iii) Set-up an integral for the total distance that the wall lizard travels. (Do not evaluate the integral.)

$$S = \int_0^2 |\vec{v}|\,dt = \int_0^2 \sqrt{1+5t^2}\,dt$$

2. [10 pts] **(Lagrange multipliers)** Find the dimensions of the rectangle of greatest perimeter that can be inscribed in the ellipse $x^2 + 2y^2 = 2$ with sides parallel to the coordinate axes.



$f(x,y) = 2(2x+2y) = 4x + 4y$

↑ to be maximized

$g(x,y) = x^2 + 2y^2 - 2 = 0$ ← constraint

$\nabla f = \lambda \nabla g$

$$\Rightarrow \begin{cases} 4 = \lambda\, 2x \\ 4 = \lambda\, 4y \end{cases} \Rightarrow 1 = \frac{x}{2y} \Rightarrow x = 2y$$

$(\lambda \neq 0, x \neq 0, y \neq 0)$

$g(x,y) = 0 \Rightarrow 4y^2 + 2y^2 = 2 \Rightarrow 6y^2 = 2 \Rightarrow y = \cancel{\pm}\frac{1}{\sqrt{3}}$ (don't accep -ive solutio

Ans.: $\boxed{2x = \frac{4}{\sqrt{3}},\ 2y = \frac{2}{\sqrt{3}}}$

3. [$4+4+4+8=20$ pts] **(Sequences and series)**

(a) Which of the following sequences converge, and which diverge? **Justify your answer.** Find the limit of each convergent sequence.

(i) $a_n = \dfrac{\sin n \cos n}{\ln(n+1)}$

let $\{b_n\} = \left\{ \dfrac{1}{\ln(n+1)} \right\}$, $\{c_n\} = \left\{ \dfrac{-1}{\ln(n+1)} \right\}$

$\forall n$, $c_n \le a_n \le b_n$ and $\lim\limits_{n\to\infty} b_n = \lim\limits_{n\to\infty} c_n = 0$

$\Rightarrow$ By squeeze thm., $\lim\limits_{n\to\infty} a_n = \boxed{0}$

(ii) $a_n = \sqrt[n]{5n^2}$

$$\lim_{n\to\infty} a_n = \underbrace{\lim_{n\to\infty} \sqrt[n]{5}}_{1} \cdot \underbrace{\lim_{n\to\infty} \left(\sqrt[n]{n}\right)^2}_{1^2} = \boxed{1}$$

(iii) $a_n = \dfrac{2^n}{n^2}$ Let $f(x) = \dfrac{2^x}{x^2}$.

$$\lim_{n\to\infty} a_n = \lim_{x\to\infty} f(x) \underset{\boxed{\text{l'Hôpital}}}{=} \lim_{x\to\infty} \frac{2^x \ln 2}{2x} \underset{\boxed{\text{l'Hôpital}}}{=} \lim_{x\to\infty} \frac{2^x (\ln 2)^2}{2} =$$

$\Rightarrow$ sequence diverges.

(b) Find the sum of the convergent series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$S_n = \left(\frac{1}{1} - \cancel{\frac{1}{2}}\right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}\right) + \left(\cancel{\frac{1}{3}} - \frac{1}{4}\right) + \dots + \left(\cancel{\frac{1}{n}} - \frac{1}{n+1}\right)$$

$$= 1 - \frac{1}{n+1}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(1 - \underbrace{\frac{1}{n+1}}_{\to 0}\right) = \boxed{1}$$

4. [$4 \times 5 = 20$ pts] **(Convergence of series)**

Which of the following series converge, and which diverge? **Justify your answer.**

(a) $\sum_{n=\cancel{1}\,2}^{\infty} \underbrace{\frac{1}{n(\ln n)^3}}_{a_n}$

Let $f(x) = \frac{1}{x(\ln x)^3}$. Then $f(n) = a_n$

f is continuous, non-negative, and decreasing on $[2, \infty)$. And

$$\int_2^{\infty} \frac{1}{x(\ln x)^3}\,dx = \int_{\ln 2}^{\infty} u^{-3}\,du = \lim_{b\to\infty}\left(\frac{u^{-2}}{-2}\Big|_{\ln 2}^{b}\right. = \lim_{b\to\infty}\left(\underbrace{\frac{1}{-2b}}_{\to 0} + \underbrace{\frac{1}{2(\ln 2)^2}}_{<\infty}\right)$$

$\boxed{u := \ln x,\ du = \frac{dx}{x}}$

$\Rightarrow \sum a_n$ converges. (by integral test)

(b) $\sum_{n=1}^{\infty} \frac{2^n\, n!\, n!}{(2n)!}$

Apply ratio test: $\frac{a_{n+1}}{a_n} = \frac{2^{\cancel{n}+1}(n+1)\cancel{!}(n+1)\cancel{!}}{(2n+2)\cancel{!}(2n+1)} \cdot \frac{\cancel{(2n)!}}{\cancel{2^n}\,\cancel{n!}\,\cancel{n!}} = \frac{2(n+1)}{(2n+1)(2n}$

$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{2n^2}{4n^2} = \frac{1}{2} < 1 \Rightarrow$ series converges.

(c) $\sum_{n=1}^{\infty} \underbrace{\cos\left(\frac{1}{n^3}\right)}_{a_n}$

$a_n \to 1 \neq 0 \Rightarrow$ series diverges (by n^{th} term test)

(d) $\sum_{n=87}^{\infty} (-1)^{n+1} \underbrace{\frac{1}{\ln n}}_{u_n} = \sum_{n=87}^{\infty} (-1)^{n+1} u_n$, where

$u_n > 0 \quad \forall n \geq 87$, u_n decreasing $\forall n \geq 87$, $u_n \to 0$.

$\Rightarrow$ series converges (by alternating series test)

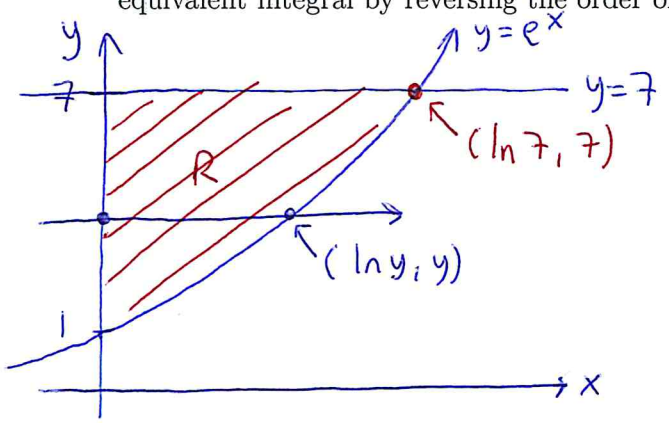
5. [6 + 6 + 8 = 20 pts] **(Double integrals)**

(a) Evaluate the iterated integral $\int_0^{\pi/2}\int_0^{\sin x} y\cos x\, dy\, dx$.

$$= \int_0^{\pi/2} (\cos x)\left(\frac{y^2}{2}\right|_0^{\sin x} dx = \int_0^{\pi/2} (\cos x)\,\frac{\sin^2 x}{2}\, dx \qquad u = \sin x \quad du = \cos x\, dx$$

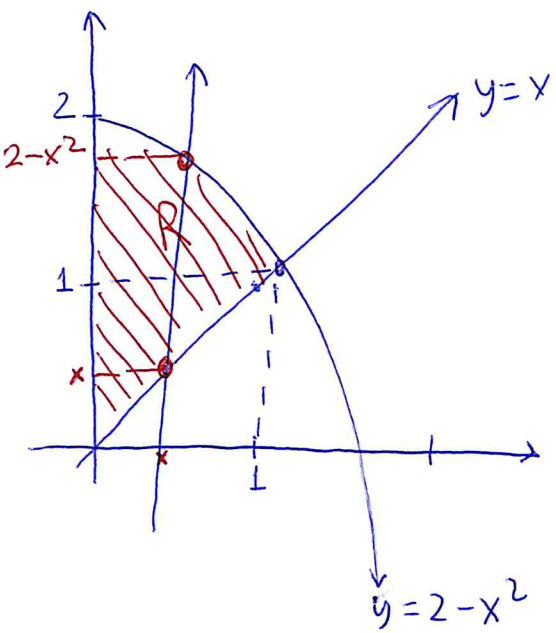
$$= \int_0^1 \frac{1}{2} u^2\, du = \frac{1}{2}\left(\frac{u^3}{3}\right|_0^1 = \boxed{\frac{1}{6}}$$

(b) Sketch the region of integration for the double integral $\int_0^{\ln 7}\int_{e^x}^{7} f(x,y)\, dy\, dx$. Set up an equivalent integral by reversing the order of integration.



$$\int_1^7 \int_0^{\ln y} f(x,y)\, dx\, dy$$

(c) By setting up a **double integral**, find the volume of the solid whose base is the region in the first quadrant of the xy-plane bounded by the curves $y = 2 - x^2$, $x = 0$, $y = x$, and whose top is bounded above by the plane $z = x$.



Find the intersection pt.:

$2 - x^2 = x \Rightarrow x^2 + x - 2 = (x+2)(x-1) =$

$x = -2$, $\boxed{x = 1}$ in 1st quadra

$$V = \int_0^1 \int_x^{2-x^2} x\, dy\, dx$$

$$= \int_0^1 x\left(y\right|_x^{2-x^2} dx = \int_0^1 x(2 - x^2 - x)\, dx$$

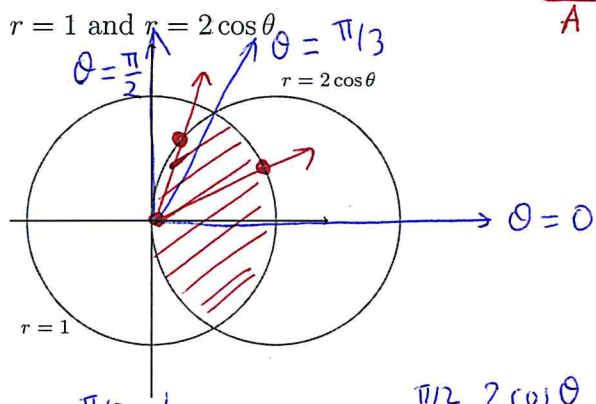
$$= \left(x^2 - \frac{x^4}{4} - \frac{x^3}{3}\right|_0^1 = 1 - \frac{1}{4} - \frac{1}{3}$$

$$= \boxed{\frac{5}{12}}$$

6. [10 + 10 = 20 pts] **(Triple integrals and integrals in polar form)**

(a) By using polar double integrals, find the area of the region that lies inside **both** circles $r = 1$ and $r = 2\cos\theta$.

A



$$A = 2\left[\int_0^{\pi/3}\int_0^1 r\,dr\,d\theta + \int_{\pi/3}^{\pi/2}\int_0^{2\cos\theta} r\,dr\,d\theta\right]$$

$$= 2\left[\int_0^{\pi/3}\left(\frac{r^2}{2}\right|_0^1 d\theta + \boxed{\int_{\pi/3}^{\pi/2}\left(\frac{r^2}{2}\right|_0^{2\cos\theta} d\theta}\right]$$

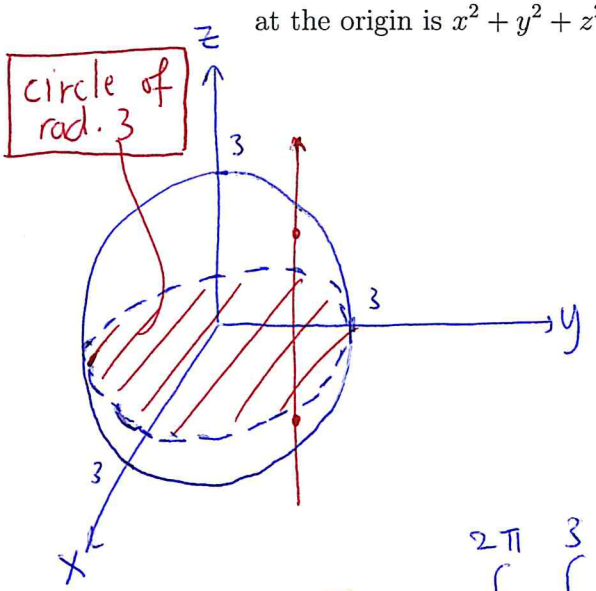
$$= \frac{4}{2}\int_{\pi/3}^{\pi/2}\cos^2\theta\,d\theta$$

$$= \cancel{2}\int_{\pi/3}^{\pi/2}\left[\frac{\cos(2\theta)+1}{\cancel{2}}\right]d$$

$$= \left(\frac{\sin(2\theta)}{2} + \theta\right|_{\pi/3}^{\pi/2}$$

$$= 2\left[\frac{\pi}{6} + \frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] = \boxed{\frac{2\pi}{3} - \frac{\sqrt{3}}{2}}$$

(b) Find the volume of a sphere with radius 3 by setting up a triple integral for its volume. Integrate with respect to z first. The equation for a sphere with radius 3 that is centered at the origin is $x^2 + y^2 + z^2 = 9$.



$$V = \iint_A \int_{z=-\sqrt{9-x^2-y^2}}^{z=\sqrt{9-x^2-y^2}} 1\,dz\,dx\,dy$$

$$= \iint_A 2\sqrt{9-x^2-y^2}\,dx\,dy$$

$$= \int_0^{2\pi}\int_0^3 2\sqrt{9-r^2}\,r\,dr\,d\theta = 2\pi\int_0^9 u^{1/2}\,du$$

convert to polar

$u = 9 - r^2$
$du = -2r\,dr$

from angular integration

$$= 2\pi\left(\frac{u^{3/2}}{3/2}\right|_0^9 = \frac{4\pi}{3}\cdot 9^{3/2} = \boxed{36\pi}$$